A Report On

Field Work

Field Work Title: Optical Fiber Cables & Connections
Location: BRB Cable Industries LTD. Kushtia, Bangladesh.

Submitted By

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PREFACE

In many cases our formal education systems build theoretical knowledge first as the basis for practical skills, however in the workplace I'd say it's often the other way round - you start doing something and at some point reach a level where you want to "fill in the gaps" in what you've learned practically and gain some sort of certification, so you have to go for theoretical-based training.

I think the best are where these are done closely together - I think practical experiential exercises alternate with theoretical sessions that underpin them. Actually knowledge is not fruitful to one unless it is a combination of both theoretical and practical in nature. So both practical and theoretical knowledge is equally important.

In our department we have a course name “Field Work” which is a good opportunity to gain knowledge about practical engineering world.

We are very honor to visit BRB Cables Industry Ltd. It is one of the largest leading cable industry in our country. We are very grateful to our honorable teacher to Associate Professor **Md Nasir Uddin Khan** and Professor **Md. Shahinuzzaman** of Islamic University, Kushtia, Bangladesh, who lead us to visit this industry. We are also thankful to authority of BRB cables industry limited.

In this year, we visit “BRB Cable Industries Ltd”, Kushtia, Bangladesh.

Group photo of 2015-2016 session

OVERVIEW ABOUT BRB CABLE INDUSTRIES

Practical and theoretical knowledge are important to fulfill one's student life. At first we have to gather knowledge which is to exact theoretically. After that, this knowledge is implemented practically. It is so that without practical work, the criteria the central theme of theoretical study is not perceivable always. There is an old saying that practical is far from theory. Actually, Knowledge is not fruitful to one unless it is a combination of both theoretical and practical in nature. So both practical and theoretical knowledge are equally important.

In our department, we have a course named "Field Work". Which is a good opportunity to gain knowledge about practical world. In this year we visit 'BRB Cable Industries Ltd.'.

11th MAY 2019, I **Md. Forhad Hossain**, along with other students of B.Sc. 2015-2016 session went to visit BRB Cable Industries Ltd. We took the permission to enter and visit the industry the day before we went there. After arriving at BRB Cables Industries, we enter into the area then the Public Relation Officer **Md. Kamruzzaman** welcomed us and gave us the briefing about the industry and warned about the rules and regulation that must be followed inside the industries.

Fig No (1): BRB Cables Industry Limited

BRB Cable Industries Ltd. a private Limited Company was established with a view to manufacture Wires & Cables in 1978. The factory started commercial production in the year 1980 and it has become a leading manufacturer of XLPE & PVC Insulated LT & HT Cables, FRLS Cables, House and Appliances Wiring Cables, Dry & Jelly filled Telecommunication Cables, Instrumentation Cables, Aluminum Overhead Conductors, Dual Coated Super Enameled Copper Wire (Winding Wire), Marine Type Cables, practically all cables required from the substation down to the lighting point. All the products are approved by BSTI (Bangladesh Standards and Testing Institution) and certified by the world-renowned, internationally reputed individual Testing laboratory CPRI (Central Power Research Institute), India.

The factory is situated in BSCIC Industrial Estate, 5 KM from Kushtia town. The factory is well equipped with modern machinery and equipment. The company has a group of qualified and experienced manpower who maintains the quality of the products. For its product quality, the company has earned fame in the country and its product has been approved and being used by BPDB, REB, DESA, DESCO, BMDA, PWD, BTMC, BSFIC, T&T, MES, BADC, Bangladesh Port Authority, Bangladesh Railway, Autonomous bodies, Private sector, Industrial and Apartment projects and individuals. To meet up increasing demand in the market, the company has set-up Unit-2 for producing Wires & Cables, AAAC, AAC & ACSR Conductor, XLPE & PVC Insulated LT & HT Cables, FRLS Cables, House and Appliances Wiring Cables, Dry & Jelly filled Telecommunication Cables,

Instrumentation Cables, Aluminium Overhead Conductors and set-up Unit-3 for producing a special type of Dual Coated Super Enameled Copper Wire (Winding Wire). The Company is marketing its products through 85 (Eighty-Five) Sales Centre. Additionally, BRB products are also sold through their sister concern, BRB Distributions Ltd. Dhaka. Besides the Sales center, the Company is marketing its products through selected Traders in all districts. The Company has a group of experienced personnel in its marketing & technical department who are well equipped to render service to its customers.

The Company was earlier certified as an ISO-9002 certified Company for its quality management system. Keeping its commitment to ensure gradual improvement of its quality products through Research and Development, prompt customer service, the company later undertook ISO-9001:2008 certificate. Moreover, the company has been awarded Bangladesh Education Scholarship Trust Gold medal, Sher-E-Bangla Gold medal, 10th Golden America Award, Arch of Europe Award, National Export Trophy (97-98) Gold and National Export Trophy (2000-2001) Bronze. The

company has been ranked 33rd amongst world leading Wires & Cables manufacturing Companies. This Study incorporated 3,000 Wires & Cables manufacturers of 200 countries.

BRB CABLES INDUSTRIES PRODUCES FOLLOWING PRODUCTS -

- **INDOOR/DOMESTIC CABLES**
- **LOW VOLTAGE CABLES**
- **MEDIUM VOLTAGE CABLES**
- **EXTRA HIGH VOLTAGE CABLES**
- **OVERHEAD CONDUCTORS**
- **SUPER ENAMELED COPPER WIRE**
- **TELECOMMUNICATION CABLES**
- **MARINE CABLES**
- **OPTICAL FIBER CABLES**
- **BRB FAN**
- **MINIATURE CIRCUIT BREAKER**

After having the short briefing about the industry he introduced us with the senior Assistant Officer Rasel, he took us inside the industry and showed us different section and gave the briefing.

INTRODUCTION

CONDUCTOR:

An electrical conductor is a substance in which electrical charge carriers, usually electrons, move easily from atom to atom with the application of voltage. Conductivity, in general, is the capacity to transmit something, such as electricity or heat. Pure elemental silver is the best electrical conductor encountered in everyday

life. Copper, steel, gold, aluminum, and brass are also good conductors. In electrical and electronic systems, all conductors comprise solid metals molded into wires or etched onto circuit boards.

HIGH TENSION CABLE:

A high-voltage cable (HV cable) is a cable used for electric power transmission at high voltage. A cable includes a conductor and insulation, and is suitable for being run underground or underwater. This is in contrast to an overhead line, which does not have insulation.

LOW TENSION CABLE:

Low tension power cable, Introduction Power Cables - An assembly of two or more electrical conductors, usually held together with an overall sheath. The assembly is used for transmission of electrical power.

OVERHEAD CABLE:

An overhead cable is a cable for the transmission of information, laid on utility poles. Overhead telephone and cable TV lines are common in North America. These poles sometimes carry overhead power lines for the supply of electric power.

UNDERGROUND CABLE:

Undergrounding is the replacement of overhead cables providing electrical power or telecommunications, with underground cables

OPTICAL FIBER:

A thin flexible fiber with a glass core through which light signals can be sent with very little loss of strength. Fiber optics transmit data in the form of light particles --

or photons -- that pulse through a fiber optic cable. The glass fiber core and the cladding each have a different refractive index that bends incoming light at a certain angle. When light signals are sent through the fiber optic cable, they reflect off the core and cladding in a series of zig-zag bounces, adhering to a process called total internal reflection. The light signals do not travel at the speed of light because of the denser glass layers, instead traveling about 30% slower than the speed of light. To renew, or boost, the signal throughout its journey, fiber optics transmission sometimes requires repeaters at distant intervals to regenerate the optical signal by converting it to an electrical signal, processing that electrical signal and retransmitting the optical signal.

CEILING FAN:

Ceiling fans are used single phase induction motor which are single phase permanent capacitor, capacitor start capacitor run and shaded pole motor. Induction motor consist of wires, winding (wound for 18, 20 or 22 poles), stator, rotor and housing.

RAW MATERIALS

COPPER:

- Copper are imported as copper cathode generally from Chili, Brazil.
- Melting point is 1083 degree Celsius.
- The diameter of a unit copper wire is 8mm.

ALUMINUM:

- Aluminum are imported as aluminum ingot generally from Chili, Brazil.
- Melting point is 673 degrees Celsius.
- The diameter of a single aluminum wire is 9.5mm.

RESIN:

- Imported from Jordan, Taiwan, Korea.
- Weight per bag is 600 to 1200 kg.

PARAFFIN OIL:

- Imported from Iran, U.A.E

WAX:

- Domestic raw material

PRODUCTION PROCESS OF WIRE:

The first machine we came across was an Extruder Machine which is used to create outer shield or insulator shield over the wire.

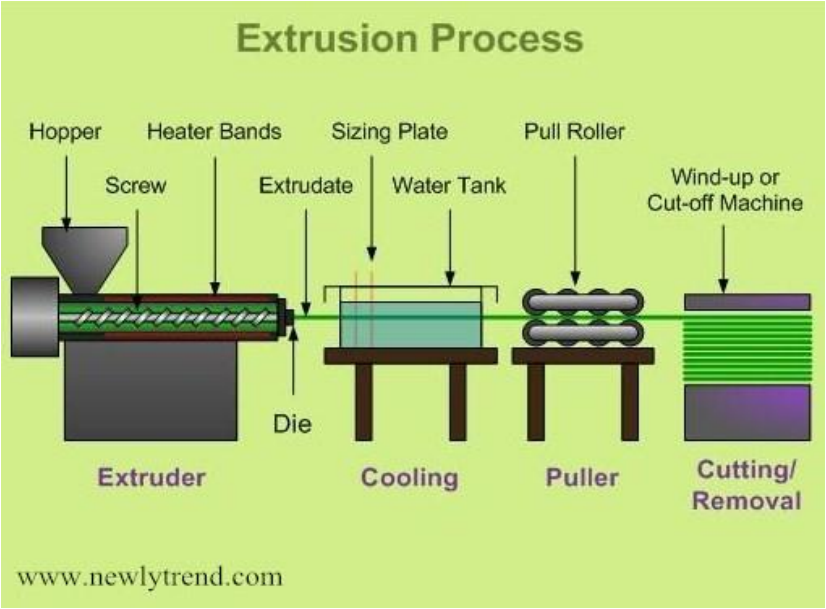


Fig No (2) :Extursion Process

Fig No (3): Extruder Machine

A printing machine which is used to print the BRB Cables name and rating over the PVC insulation shield after one-meter distance using Infrared Ray. Wires enter from one end, come across out printed from other end.

Fig No (4): Printed cable

Drawing Machine which is produced wire of different diameters from 1cm to 1 mm applying pressure.

Fig No (5): Drawing Machine

Multi wire drawing which takes multiple wires, say 7 wires as input and produces 7 wires as output. Meanwhile it reduces the diameter of the wire, at the same time imulsioning it and enameling it through heating. It produces very thin wire having diameter of .19 mm.

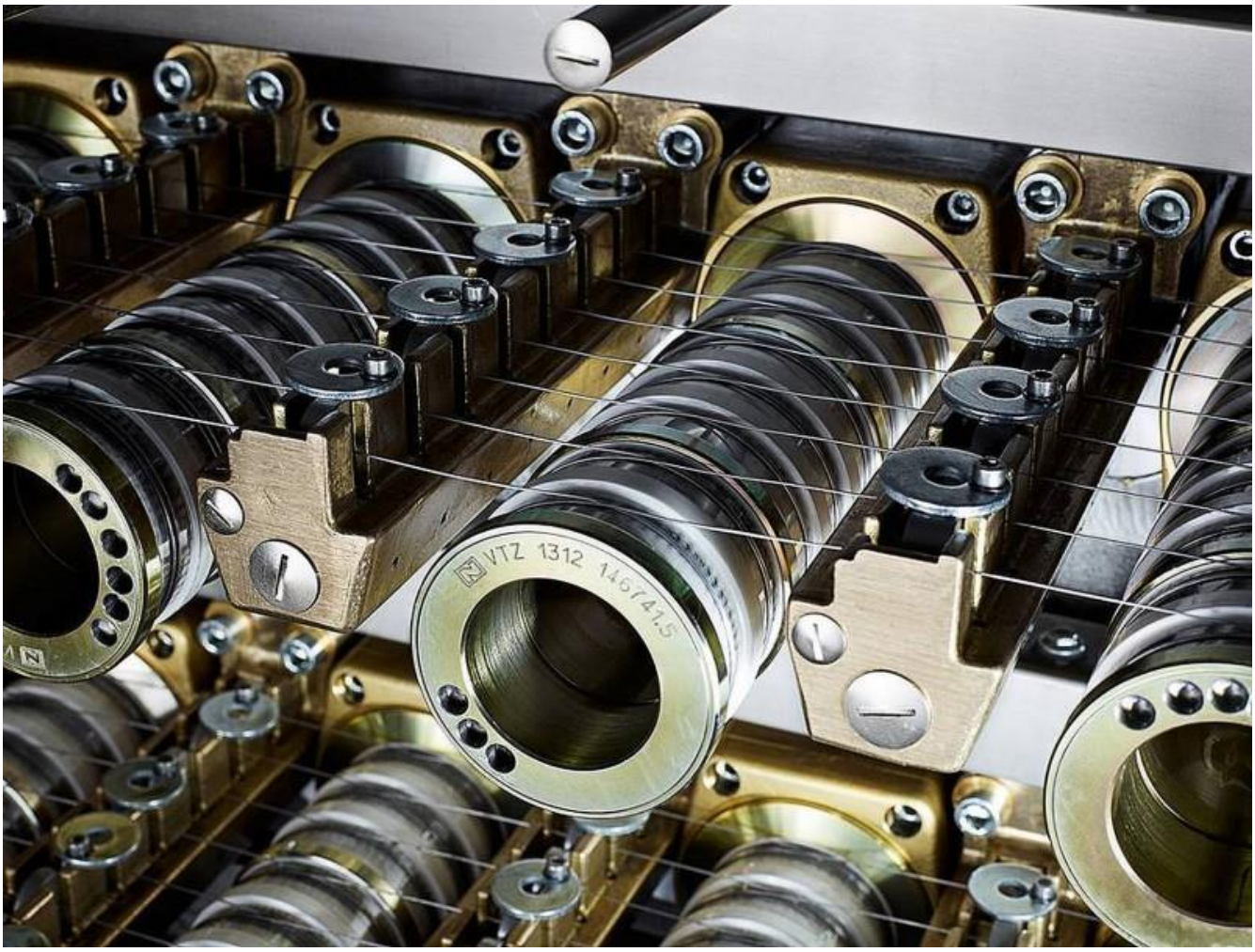


Fig No (6): Multi wire Drawing Machine

Twisting Machine which can be used to twist 61 copper wire or cable together. It was AEI 61 BOBBIN (Rigid).

Fig No (7): Twisting machine

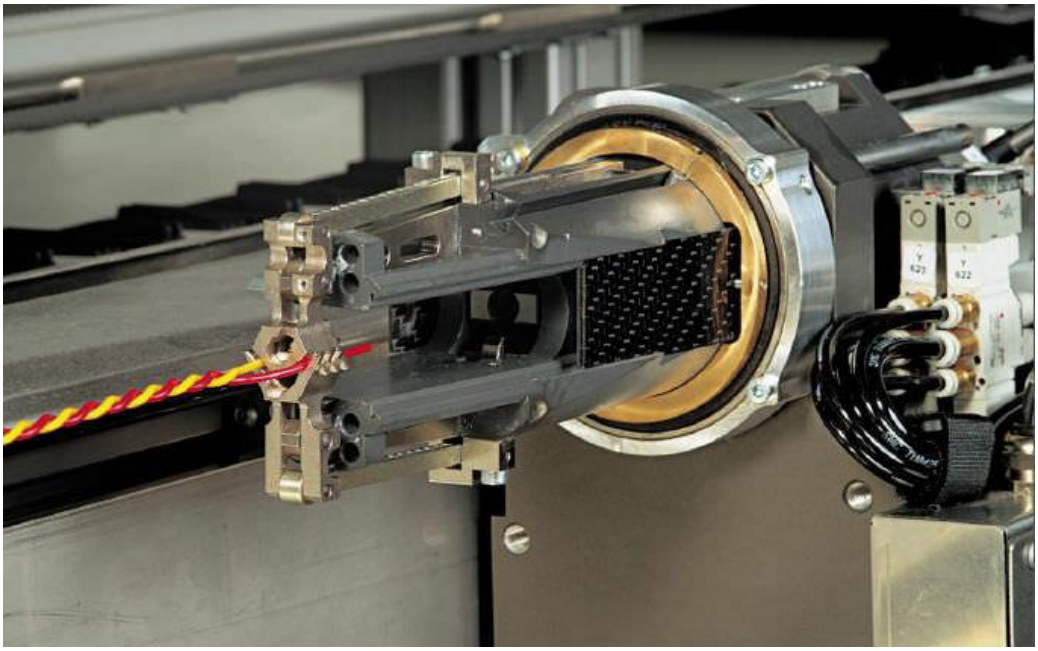


Fig No (8): Twisting Machine

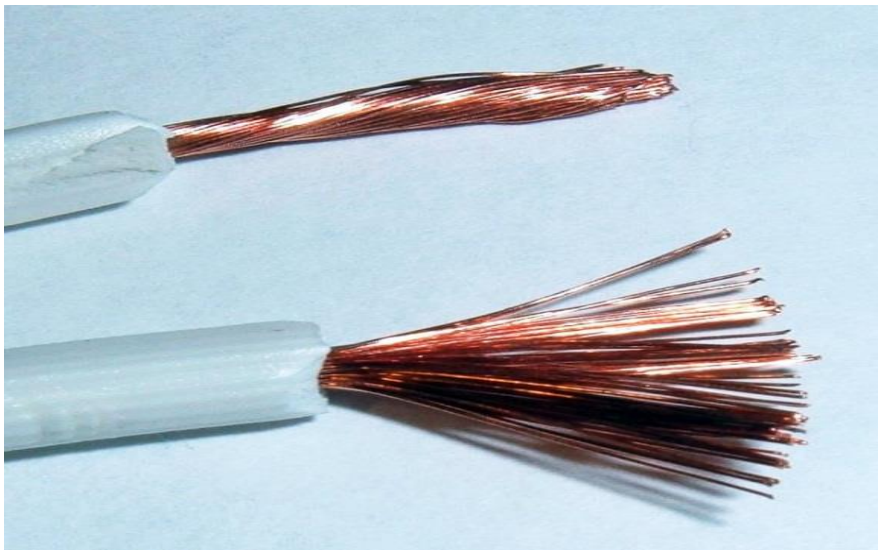


Fig No (9): twisted and non-twisted wire

The copper depth machine which also known as Brunching Machine is also used for twisting purpose.

High tension cable where the leakage current is reduced using semiconductor insulation. It is rated as 33 KV.

Fig No (10): High Tension cables produced by BRB

CCP AND Sioplas Machine which are both used for thick cable insulation.

The generators which supply the necessary power during power failure.

Extruder Machine which produces the PVC used for isolation over the copper wire.

Fig No (11): PVC Extruder Machine

The Machine which produces Super Enamled wire known as Varnish Insulation Machine by creating a layer of varnish over the bare cable. After enameling the wires are taken over various reels, and then wires of different thickness are produced using different drawing machines.

CABLE INFORMATION:

The role of cable is to carry an audio or video signal from one device to another. Cables carry signals between DVD players and TVs, stereo receivers and speakers, and computers and video projectors. Cables don't change the nature of the audio or video signal they carry. They don't convert or process signals in any way.

That's the job of the devices on either end. The cable itself is just the messenger. The cables are made up of three basic parts: conductor, shielding and connector. The

conductor is the wire that actually carries the signal. One or more layers of shielding prevent the wire from acting as an antenna that picks up radio frequency interference (RFI) and electromagnetic interference (EMI) . The connector is the plug at the end of the cable that connects to your device.

Cables are important components of any home stereo or home theater setup. If you don't use the right cables for the right job, then you could end up with subpar sound or picture quality. And if you use cables that are damaged or otherwise junky, you could really end up with a lousy experience. For audiophiles, using low-end cables with a $5,000 plasma HDTV is like putting crappy tires on a Ferrari.

Some audio/video experts argue that consumers should spend 20 percent of their total system cost on cables alone. The truth is that many cheaper cables offer a listening/viewing experience that most consumers could never tell from the really high-end cables.

To keep things simple, we're going to break down the overwhelming amount of cable types into three categories: audio-only, video-only and audio/video cables that carry both sound and picture.

Types of Cables in BRB Industry:

There are various kinds of cables in BRB Industry.

Such as:

- Domestic Cables
- L.T(Low Tension) Cables
- H.T(High Tension) Cables
- Marine Cables

- Co-axial Cables
- Telephone Cables
- Overhead Cables

Describe about this cables:

DOMESTIC CABLES:

Multiple copper wire conductor with twisting, PVC insulation sheathing. Its capability ranges 200 to 750 volts. Domestic cables are mainly used as house wire. Some types are- BYA, NYY, BYMY, BYEY etc.

Type: BYA

Standard: BDS-900 and BS-6004

Voltage:450/750 volts

Fig No (12): Domestic cable

CONSTRUCTION:

Plain annealed Copper Conductor, PVC insulated, non-sheathed single core cable. Suitable for use in surface mounted or embedded conduits or trucking. Also suitable for field protected installation in lighting fittings and inside appliances.

Type: BYFY

Standard: BDS-900 and BS-6004

Voltage:300/500 volts

Fig No (13): Domestic cable

Construction: Plain annealed Copper Conductor, PVC insulated, Two or Three cores laid FLAT and PVC sheathed overall. Suitable for use in Fixed installation in dry or damp premises and for installation Walls or Boards and in Channels or embedded in plaster. Color of Sheath: GREY/WHITE

Type: NYY

Standard: BDS IEC-60502 and VDE-0271/3.69

Voltage:600/750 volts

Fig No (13): Domestic cable

Construction: Plain annealed Copper Conductor, PVC insulated, PVC sheathed, Single core cable. Suitable for using indoors, outdoors, underground and in water for continuous permissible service of 720/1200 volts. Color of Sheath: BLACK.

L.T (LOW TENSION) CABLE:

L.T (low tension) cable are made by following sections:

1. Multiple copper with twisting
2. Semiconductor
3. XLPE (Cross link poly ethylene)
4. Semiconductor
5. Copper tape
6. Aluminium steel wire
7. Sheathing PVC.

Its capability ranges 750 to 11 kilovolts. L.T(Low tension) Cables are mainly used as Industry/ Large buildings. Some types of L.T(Low tension) cable are- 2XSEY,2XFGY etc.

Type: 2XFGY

Standard: IEC-60502-1

Voltage:600/1000 volts

Fig No (12): Low tension cable

Construction: Plain annealed Copper, XLPE Insulated, Three, Three & Half or Four cores laid up, PVC Common covering, Flat, Galv. Steel wires armoring with helical steel tape binder & PVC Sheathed overall.

Suitable for use in indoors, outdoors, underground and in water for continuous permissible service voltage of 720/1200 volts.

Color of Sheath: BLACK

H.T(HIGH TENSION) CABLE:

H.T (High tension) cable are made by following sections:

1. Multiple copper with twisting
2. Semiconductor
3. XLPE (Cross link poly ethylene)
4. Semiconductor
5. Copper tape
6. Aluminium steel wire
7. Sheathing PVC.

High-voltage cables have the structural elements of one or more conductors, insulation, and a protective jacket. High-voltage cables differ from lower-voltage cables in that they have additional internal layers in the insulation jacket to control the electric field around the conductor. Its capability ranges up to 11 kilovolts. H.T(High tension) Cables are mainly used as Industry. Some types of H.T(High tension) cable are-2XFGBY,2XRGBY,2XHSY etc.

Type: 2XHSY

Specification: IEC 60502-2

Uo/U (Um)=3.6/6 (7.2) kv.

Permissible Service Voltage:3.8/6.5 kv.

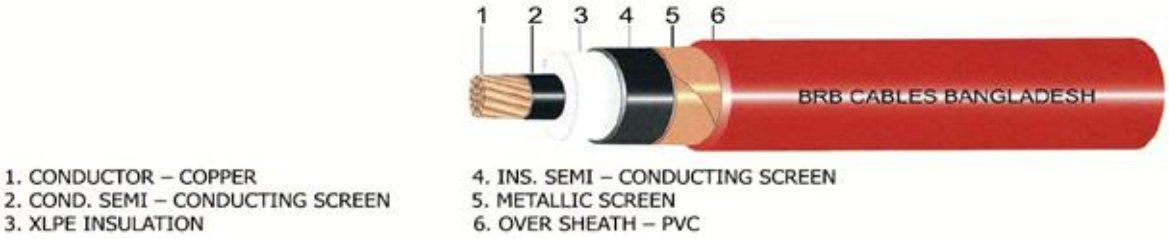


Fig No (12): High tension cable

MARINE CABLE:

Marine cable is made by following sections:

1. Copper wire.
2. Insulation with PVC/XLPE.
3. Inner Insulation PVC

4. Copper braiding.

5. Outer sheathing with PVC.

Marine Cables are mainly used as Ship board. Some types of Marine Cables are-Mc-210Q,

MC-210T etc.

MC – 210Q (Cu or Coated Cu / XLPE / PVC / SWB / PVC)

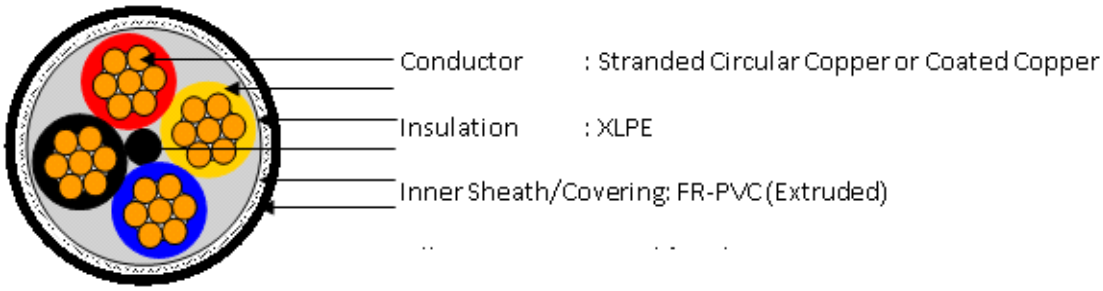


Fig No (13): Marine cable

Cross-Section Diagram of Stranded Circular Conductor MC – 210Q

(With wire braided, Armored Double Sheathed Cable)

APPLICATION

This cable is suitable for use in fixed installations for power supply in commercial ships. (For lighting control and power)

CO-AXIAL CABLE:

Co-axial cable is made by following sections:

1. Single Copper wire.

2. Insulation MDPE (Medium density polyethylene) or LDPE (Medium density polyethylene).

Co-axial cables are mainly used as dish cable. Some types of co-axial cable are RG-6, RG-5, MIL-c-17 etc.

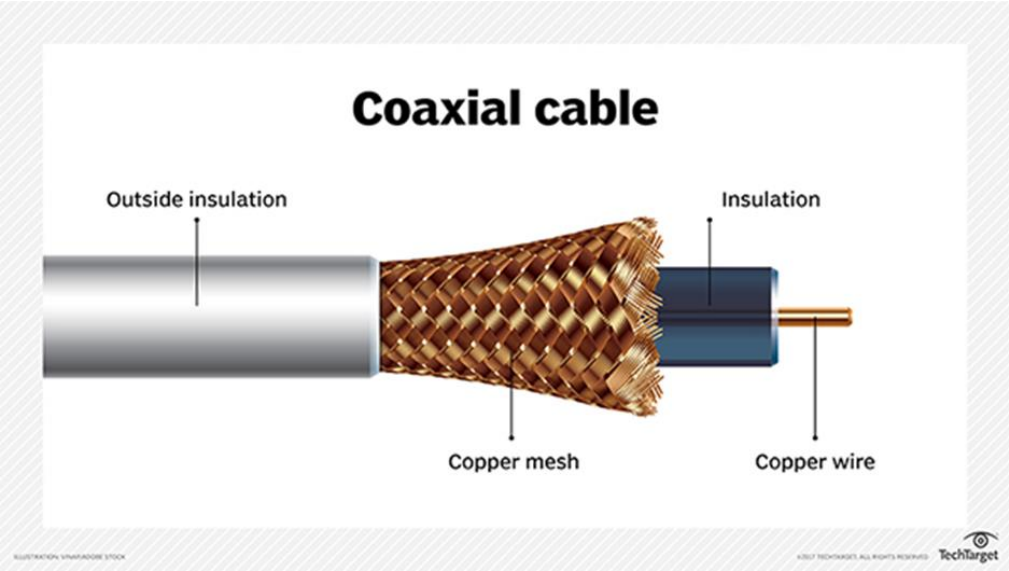


Fig No (14): Co-axial cable

Type: CO-AXIAL CABLES

Standard: MIL-C-17

Fig No (15): Coaxial cable

Construction:

Plain annealed Copper conductor, Polythene insulated, Copper wires braided and sheathed overall. Suitable for use as aerial down leads in local and fringe reception areas at both very high frequency and ultra-high frequency.

TELEPHONE CABLE:

Telephone cable are made by following sections:

1. Single Copper wire.
2. Insulation LDPE (Medium density polyethylene).
3. Twisting Core.
4. Outer Sheathing with PVC

OVER HEAD CABLE:

Over Head cable are made by following sections:

1. Bear with Copper / Aluminum.
2. Insulation with PVC.

Over Head cables are mainly used to connect the transformer from house. Some types of Over Head cable are ABC Cable, D-7, D-11etc.It also use in PDB, DPDC, WASA.

We visited three section of BRB cable industry limited, Kushtia, Bangladesh.

There are-

1. Production section.
2. Compound section.

Describe about these two sections:

1)PRODUCTION SECTION:

We have visited this section some machines. This machines are:

1. Drawing machine.
2. Bunching machine.
3. Extruder machine.

Describe about these types of machines:

1.DRAWING MACHINE:

Draw the 8 mm copper /Al rod and to make various size of wire i.e 2.56mm,1.73mm,1.4mm,1.2mm,0.94mm,0.8mm,0.4mm,0.3mm.When pressure is applied to the Cu/Al rod, then it's length is increased and diameter decreased. The main function of this machine is to provide flexibility of Cu/Al wire through the annealing process.

2.BUNCHING /TWISTING MACHINE:

The main function of this machine is to twist the multiple Cu/Al wire. It also twists core.

3.EXTRUDER MACHINE:

It insulates the Cu/Al wire. It uses PVC (Poly Vinyl Chloride) and XLPE (Cross link poly ethylene) for insulation. The insulation is done for mechanical protection and safety of Cu/Al wire.

ADVANTAGES OF PVC INSULATION:

PVC is the most widely used polymer for cables production in Europe. It is mainly dominant in the low voltage and some specialist applications. Telecommunication is

also an important application for PVC. PVC cables have a number of benefits, such as:

- Good electrical and insulation properties over a wide temperature range
- Inherent fire safety
- Excellent durability and long-life expectancy
- Easy processing characteristics to achieve desired specification for end-products
- Cost-effectiveness
- Recyclables - no cross-linking therefore the ability to be reprocessed back into cable applications, see further down
- Compares favorably to alternative materials using LCA methodologies primarily due to lower usage of non-renewable resources i.e. 43% derived from oil/gas and 57% derived from salt
- Highly resistant to degradation by ultra violet light
- Cheap Highly resistant to degradation by ultra violet light
- Cheap

ADVANTAGES OF XLPE INSULATION:

The use of XLPE as the insulation for HV and EHV transmission cables has grown steadily since the early 1990s. Today XLPE insulation is the system that is preferred

to traditional lapped insulation (paper or paper polypropylene laminate) fluid-filled cables. This preference stems from the high reliability, low dielectric losses and low environmental impact that can be achieved with

XLPE. Consequently, XLPE cables are now used at high electrical stresses in a

Growing number of high voltage (36-161 kV, IEC60840) applications.

COMPOUND SECTION:

There are two sections.

- PVC Compounds – Various Grades:

We are a leading manufacturer of premium PVC cable compounds for wide application in the cable industry. Our range of compounds is available for numerous applications for Power Cables, Control and Instrumentation Cables, Winding wire for submersible Pump Motor, Three Core Flat Cables, Appliance Cords, etc. Our PVC compounds are available for both insulation and sheathing.

These compounds are also used for House Wiring, Telecom Cables, Automotive Cables, etc. All our products & PVC compounds conform to standards as per ISO standards. Our compounds are manufactured from top quality Color Master-batches & are weather resistant, UV resistant, non-migrant, non-blooming.

XLPE COMPOUND:

This company specializes in color resin compounds, master batches, tailor-made pigments, and dry blend for boulders. Dry blends are economical than moulding powders and pellets and hence has great demand. The company offers tailor-made dry blends as per customer requirements. The resin compound produced by this

company is cross linkable polyethylene compound (XLPE) for 6.6 KV low voltage power cable and 35KV overhead cable. This company markets its Cross linkable polyethylene compound under Plaslink brand with two categories of compound which is ambient cured and hot water/steam cured.

OPTICAL FIBER

In this section we observed how to produce various kinds of telecommunication Such as Optical fiber ,UTP cables(CAT5 and CAT6),Co-axial cable.

OPTICAL FIBER CABLE :

An optical fiber is a flexible, transparent fiber made of glass(silica) or plastic, slightly thicker than a human hair. It functions as waveguide or light pipe to transmit light between the two ends of the fiber. The field of applied science and electrical engineering concerned with the design and application of optical fibers is known as fiber optics. Optical fibers are widely used in fiber-optic communication, which permits transmission over longer distances and at higher bandwidths than other forms of communication. Fiber are used instead of metal wires because signals travel along them with less loss and are also immune to electromagnetic interference. Fibers are also used for illumination and are wrapped in bundles so that may be used to carry images, thus allowing viewing in confined spaces. Specially designed fibers are used for a variety of other applications including sensors and fiber lasers.

An optical fiber cable is a cable containing one or more optical fibers that are used to carry light. The optical fiber elements are typically individually coated with plastic layers and contained in a protective tube suitable for the environment

where the cable will be deployed. Different types of cable are used for different applications, for example long distance telecommunication, or providing a high speed data connection between different parts of a building.

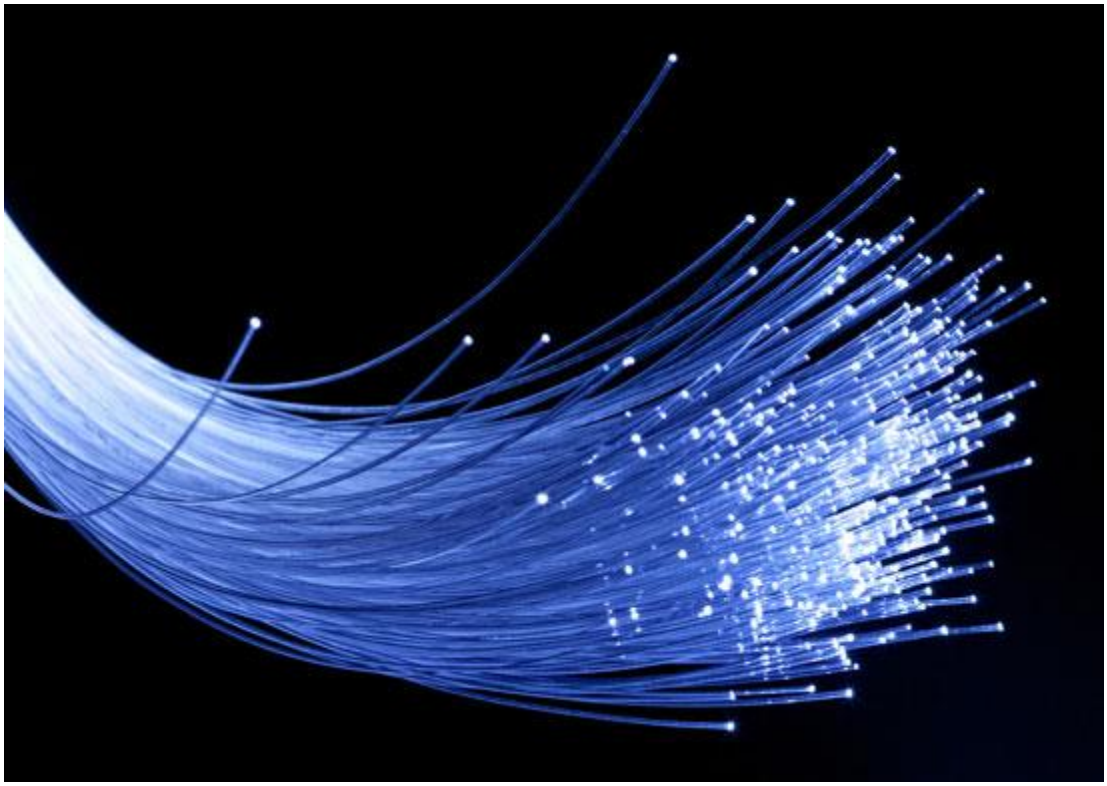


Fig No (16): Optical fiber cable

Optical fiber consists of a core and a cladding layer, selected for total internal reflection due to the difference in the refractive index between the two. In practical fibers, the cladding is usually coated with a layer of acrylate polymer or polyimide. This coating protects the fiber from damage but does not contribute to its optical waveguide properties. Individual coated fibers (or fibers formed into ribbons or bundles) then have a tough resin buffer layer and/or core tube(s) extruded around them to form the cable core. Several layers of protective sheathing, depending on

the application, are added to form the cable. Rigid fiber assemblies sometimes put light-absorbing ("dark") glass between the fibers, to prevent light that leaks out of one fiber from entering another. This reduces cross-talk between the fibers, or reduces flare in fiber bundle imaging applications.

OPTICAL FIBER CONSTRUCTION:

Bare Solid Copper Conductor, PE Insulated and PVC sheathed overall. Suitable for use in transmitting telecommunication signals. Normally they are suitable for expose or concealed installation on the walls. Permissible voltage 200 V.

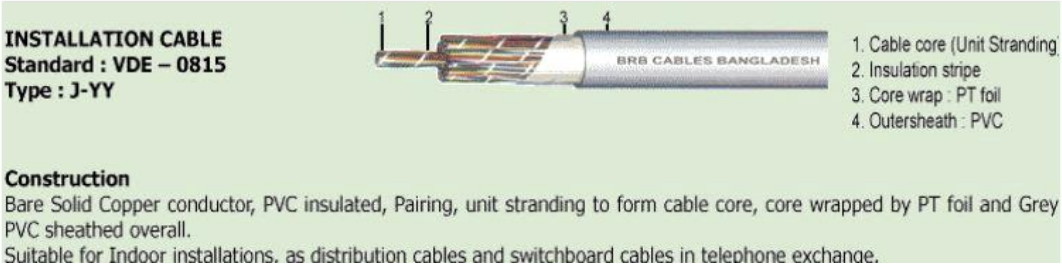


Fig No (17): Installation cable

Manufacturing Products:

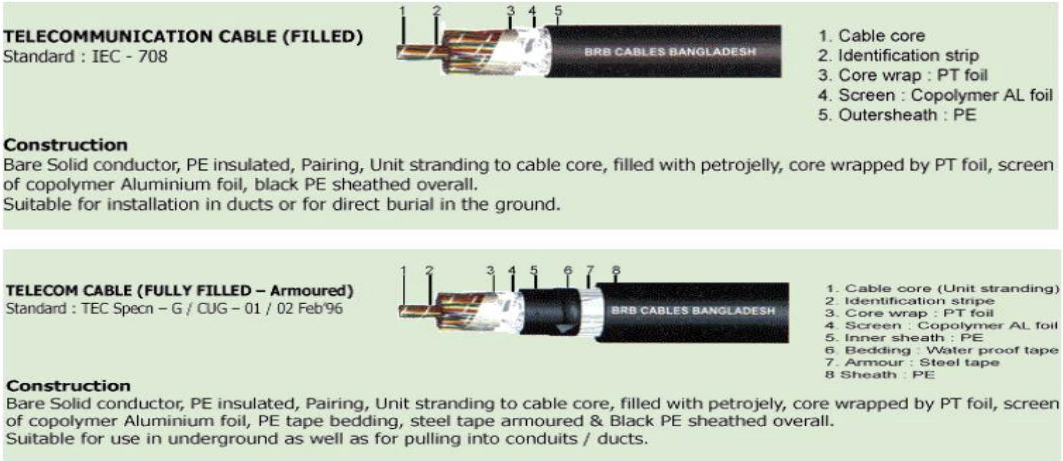


Fig No (18): Tele communication Cable

LAB TEST OF OPTICAL FIBER:

IMPACT TEST:

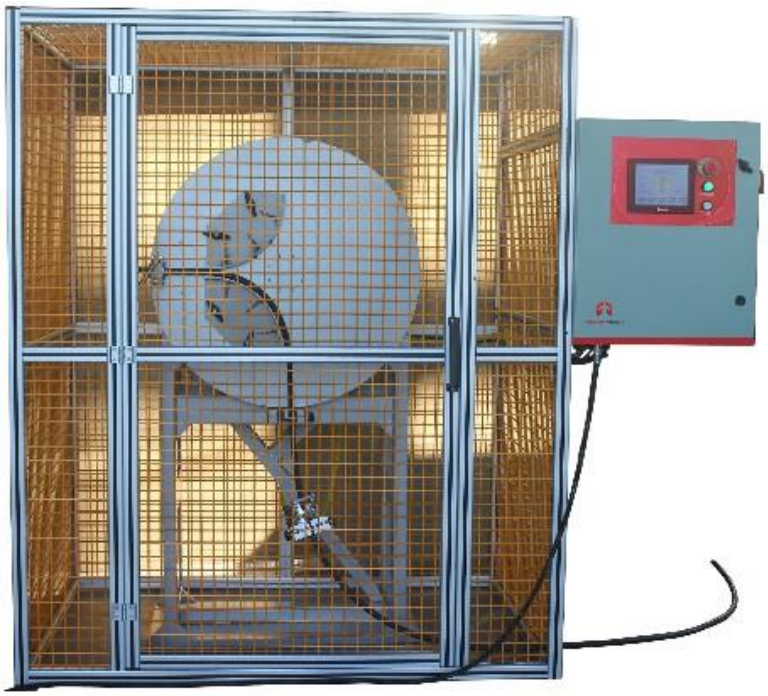
Torontech's Optical Fiber Cable Impact Testing Machine is intended to determine the ability of optical fiber cable withstand impact, it has a hammer with a 20mm diameter to drop vertically on a cable sample fixed on a flat steel plate. It also allows a single or multiple repeated impacts to be imparted on the cable sample. The energy of the impact is determined by the drop height and by the weight of the hammer.

Fig: Impact testing machine

BENDING TEST:

Optical Fiber Cable Repeated Bending Testing Machine is used to determine the ability of a fiber optic cable to withstand repeated bending (cyclic flexing).

Optical Fiber Cable Impact Testing Machine TTOFCRB Series allows a cable sample to be bent backwards and forwards at an angle of 180° by bending extreme ends of cables at 90° in opposite directions, while the sample is subjected to a tensile load.



The bending arm is designed to hold the cable securely during the entire test, without crushing the optical fibers or inducing optical loss. The bending radius is controlled by replaceable cushioning reels, chosen to match the cable bending radius. The machine is capable of cycling at a rate of 30 cycles per minute.

TWIST TEST:

Optical Fiber Cable Torsion Testing Machine is used to determine the ability of optical fiber cable to withstand mechanical twisting. This machine is specially designed in accordance with IEC-60794-1-2-E7

Fig. Twist test machine

UTP CABLES:

The UTP means unshielded twisted pair cable. There are produced two types of UTP cables namely cat5 and cat6 cables.

Fig No (19) UTP cables

Although these cables are used for a variety of purposes, they are most commonly used as network cables, i.e. They connect computers and servers to modems and

(ultimately) is ps. Like virtually everything else in the technology sector, these cables have undergone significant development over the years, and the newest cables are capable of significantly greater performance than previous cables. The difference between a Cat5 vs Cat6 cable is not only higher speeds but reduced “cross talk”. It is worth noting that both Cat5 vs Cat6 cables utilize the same end piece, i.e. they can “plug in” to the same ports. The differences between each of these cables are in their capabilities, as well as the methods and materials used to create them. The “end” that all the cables have in common is known as RJ-45, and it is capable of plugging into any Ethernet jack on a computer, router, or another similar device. Nobody in the industry expects this to change anytime soon.

CAT5 CABLE:

Cat5 cable is broken into two separate categories: Cat5 and Cat5E cables. Cat5 has become obsolete in recent years, due to its limitations compared to Cat5E and Cat6 cables. Although the Cat5 cable can handle up to 10/100 Mbps at a 100mhz bandwidth (which was once considered quite efficient), the newer versions of Cat cables are significantly faster.

Cat5E cable (which stands for “Cat5 Enhanced”) became the standard cable about 15 years ago and offers significantly improved performance over the old Cat5 cable, including up to 10 times faster speeds and a significantly greater ability to traverse distances without being impacted by crosstalk.

CAT6 CABLE:

Cat6 cables have been around for only a few years less than Cat5E cables. However, they have primarily been used as the backbone to networks, instead of being run to workstations themselves. The reason for this (beyond cost) is the fact that, while Cat6 cables can handle up to 10 Gigabits of data, that bandwidth is limited to 164 feet — anything beyond that will rapidly decay to only 1 Gigabit (the same as Cat5E).

Cat6A is the newest iteration and utilizes an exceptionally thick plastic casing that helps further reduce crosstalk. The biggest distinguishing difference between Cat6 and Cat6A cables is that Cat6A can maintain 10 Gigabit speeds for the full 328 feet of Ethernet cable.

Ultimately, those who want to have the most “future proofed” cable will want to go with Cat6A. However, for most resident and commercial purposes, Cat5E and Cat6 cables should be more than sufficient.

CO-AXIAL CABLE:

Coaxial cable or coax is a type of cable that has an inner conductor surrounded by a tubular insulating layer, surrounded by a tubular conducting shield. Many coaxial cables also have an insulating outer sheath or jacket.

Fig No (20): Co axial cable

Coaxial cables are a type of cable that is used by cable TV and that is common for data communications. Taking a a round cross-section of the cable, one would find a single center solid wire symmetrically surrounded by a braided or foil conductor. Between the center wire and foil is a insulating dielectric.

LOVELY CEILING FAN:

The name of BRB'S ceiling fan is lovely fan. A ceiling fan is a mechanical fan, usually electrically powered, suspended from the ceiling of a room that uses hub-mounted rotating paddles to circulate air.

A ceiling fan rotates much more slowly than an electric desk fan; it cools people effectively by introducing slow movement into the otherwise still, hot air of a room. Fans never actually cool air, unlike air-conditioning equipment, but use significantly less power (cooling air is thermodynamically expensive). Conversely, a ceiling fan can also be used to reduce the stratification of warm air in a room by forcing it down to affect both occupants' sensations and thermostat readings, thereby improving climate control energy efficiency.

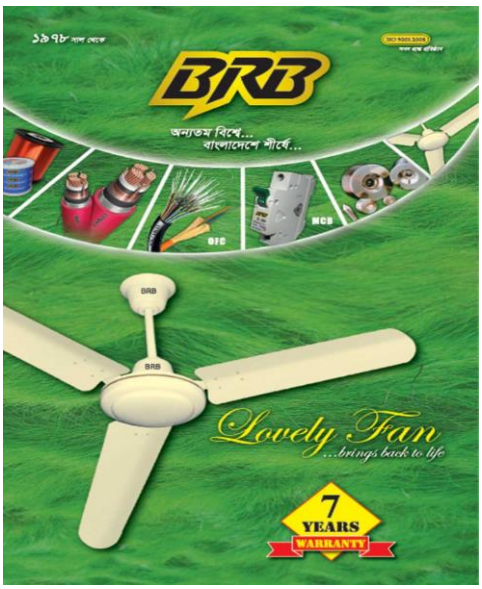


Fig: Lovely Fan

ARMATURE:

Fig No (22): Armature

Fig No (23): Stator and Rotor

Ceiling Fan Coil Winding Machine is suitable for winding ceiling fan stator, external armature such as frequency conversion alternator and generator armature. In most generators, the field magnet is rotating, and part of the rotor, while the armature is stationary, and is part of the stator. Both motors and generators can be built either with a stationary armature and a rotating field or a rotating armature and a stationary

field. The pole piece of a permanent magnet or electromagnet and the moving, iron part of a solenoid, especially if the latter acts as a switch or relay, may also be referred to as armatures.

FEATURES:

This ceiling fan coil winding machine has two working stations. Controlled by microcomputer, it can do auto winding, auto array, auto indexing; array and indexing are controlled by stepping motor, winding is controlled by frequency.

Fig No (24): converting motor, turn number could be accurate up to 0.1 rounds

ROTOR:

The rotor is a moving component of an electromagnetic system in the electric

Fig No (25): Rotor

motor, electric generator, or alternator. Its rotation is due to the interaction between the windings and magnetic fields which produces a torque around the rotor's axis.

TYPE AND CONSTRUCTION OF ROTORS:

Induction (asynchronous) motors, generators and alternators (synchronous) have an electromagnetic system consisting of a stator and rotor. There are two designs for the rotor in an induction motor: squirrel cage and wound. In generators and alternators, the rotor designs are salient pole or cylindrical.

OPERATING PRINCIPLE:

Fig No (26): winding machine **(**Wound the coil in the stator within 2.5 minutes. (Automatically)

In a three-phase induction machine, alternating current supplied to the stator windings energizes it to create a rotating magnetic flux. The flux generates a magnetic field in the air gap between the stator and the rotor and induces a voltage which produces current through the rotor bars. The rotor circuit is shorted and current flows in the rotor conductors. The action of the rotating flux and the current produces a force that generates a torque to start the motor. An alternator rotor is

made up of a wire coil enveloped around an iron core. The magnetic component of the rotor is made from steel laminations to aid stamping conductor slots to specific shapes and sizes. As currents travel through the wire coil a magnetic field is created around the core, which is referred to as field current The field current strength controls the power level of the magnetic field.

Direct current (DC) drives the field current in one direction, and is delivered to the wire coil by a set of brushes and slip rings. Like any magnet, the magnetic field produced has a north and a south pole. The normal clockwise direction of the motor that the rotor is powering can be manipulated by using the magnets and magnetic fields installed in the design of the rotor, allowing the motor to run in reverse or counter clockwise.

BLADES:

In a typical fan there are three blades. But in modern day fan there are more blades according to the air required. All the blades have a particular structure so that the air can be sucked properly and delivered to the desired place with satisfactory level.

Fig No (27): Blades

SUPER ENAMELED COPPER WIRE:

Fig No (28): Super Enameled wire

 Magnet wire or enameled wire is a copper or aluminium wire coated with a very thin layer of insulation, generally by polyamide.

 The wire itself is most often fully annealed, electrolytic ally refined copper

 Polyimide insulated magnet wire is capable of operation at up to 250 °C.

 Diameter of a single core wire is up to 8mm.

 Coating is also known as varnish.

 It is used in the construction of transformers, inductors, motors, speakers,

Hard disk head actuators, electromagnets, and other applications that

Require tight coils of insulated wire.

 BRB uses two types of machine horizontal and vertical.

There are three others section namely miniature circuit breaker section (MCB), extra high voltage section and quality control department.

MINIATURE CIRCUIT BREAKER SECTION (MCB):

In this section miniature circuit breakers are produced. In this section the most workers are women and 100 women work and that's why we were not permitted to visit this section.

QUALITY CONTROL:

Cables and Conductors are manufactured here strictly according to the international standard specification. This stamp of quality is ensured by the facts that the cables are being manufactured under license from the world-renowned Kabel-werke-reinshagen of Germany. It is a matter of great pride that the fruits of accumulated experience of Kabel-werke-reinshagenenriched by its constant research and development work over a period of hundred years is being shared by BRB Cables Limited from its very inception. Quality control laboratories and modern PVC compounding plant ensure the physical, chemical and electrical properties of all products as per standard specification. PVC cables are manufactured in Eastern Cables Limited mainly according to Bangladesh Standard (BDS), German Metric Standard (VDE), and British Metric Standard. Bare and insulated AAC and ACSR conductors are also manufactured here according to B.S and ASTM. On demand, cables and conductors are also manufactured here according to any International Standard.

CONCLUSION:

The field work is not only an academic course. It is also lesson how to be practical man. By visiting BRB we must use our skill of academic knowledge into practical. It is an achievement to all of us. The field work is a part of our syllabus for completion of third year. The field is also very helpful for our future life. In this field work we observed that total system of BRB cable industries. We also got an idea about cables and its production technique, the total system is very complicated and secured. We also got a concept about the engineer's role in this organization. We also know that cables are used to transfer large amount of voltage and current with safety. We beg apology for the unavoidable mistake that might happen in this report.

Finally, a special thanks to our honorable teacher to arrange such as place for our field work.

I also wish to thank to our classmates who assist our honorable teacher during our tour.